Foundations of the human body

Practice Paper 1 | 30 marks | Suggested time: 35 minutes

Name: _____ Class: _____

Instructions: Answer ALL questions. Use full sentences when asked to explain. Show calculations and units. The number in square brackets tells you the marks available.

Question 1 Aerobic respiration — 7 marks

a) State the word equation for aerobic respiration. **[2]**

b) Give TWO different uses of the energy released by aerobic respiration. **[2]**

c) Explain how glucose from a meal and oxygen from the lungs reach a muscle cell. **[3]**

Question 2 Digestion and nutrients — 8 marks

a) State ONE function of each of these organs: mouth, oesophagus, stomach, small intestine. **[4]**

b) State ONE main function of each nutrient: protein, lipid, fibre. [3]

c) Why must many large food molecules be digested before they can be absorbed? [1]

Question 3 Digestion and absorption — 8 marks

a) Explain the difference between physical and chemical digestion. [2]

b) Explain TWO ways the structure of villi makes nutrient absorption efficient. [4]

c) Define diffusion, including the direction of net particle movement. [2]

Question 4 Gas exchange — 7 marks

a) Name the tiny air sacs where most gas exchange occurs in the lungs. **[1]**

b) State the direction in which oxygen and carbon dioxide diffuse between alveoli and capillary blood. **[2]**

c) Explain THREE features that help alveoli exchange gases efficiently. **[3]**

d) Why is oxygen entering the blood important for cells? **[1]**

End of paper. Check that you have answered all parts. Total: 30 marks.